Appendix A

CompTIA Network+ N10-009 Certification Exam Objectives

This text covers material related to all the examination objectives for the CompTIA Network+ exam N10-009, which was released by CompTIA (the Computing Technology Industry Association) in 2024. The official list of objectives is available at CompTIA's website, comptia.org. For your reference, the following tables list each exam objective and the chapter and section of this course that explains the objective, plus the amount of the exam that will cover each certification domain. Each objective belongs to one of five domains (or main categories) of networking expertise. For example, the task of comparing and contrasting different 802.11 standards belongs to Objective 1.5 in the "Networking Concepts" domain, which altogether accounts for 23 percent of the exam's content.

Domain	Percentage of examination
1.0 Networking Concepts	23%
2.0 Network Implementation	20%
3.0 Network Operations	19%
4.0 Network Security	14%
5.0 Network Troubleshooting	24%
Total	**100%**

As you read through the exam objectives, pay close attention to the verbs used in each objective, such as "Explain" and "Summarize." These words indicate how deeply you should know the content listed.

For example, an objective that says, "Explain the purposes ..." or "Compare and contrast ..." expects you to understand the concepts listed, be able to identify those concepts in a scenario, and answer questions about the concepts. These objectives are identified at the "Understand" level of Bloom's Taxonomy, which is a theoretical model used to distinguish between different kinds of learning. However, an objective that says, "Given a scenario, use ..." or "Given a scenario, implement ..." expects you to be able to put those concepts to work, which is the "Apply" level in Bloom's Taxonomy.

Any objective that begins with the words "Given a scenario" is more likely to show up on the exam as a performance-based question or with details from a scenario rather than simply as a basic multiple-choice question. Additionally, some objectives begin with "Given a scenario, troubleshoot...." Due to the more advanced knowledge and skill required to troubleshoot a problem, these objectives are identified at the "Analyze" level of Bloom's Taxonomy.

Domain 1.0 Networking Concepts—23% of Examination

Objective	Section	Bloom's Taxonomy
1.1 Explain concepts related to the Open Systems Interconnection (OSI) reference model. • Layer 1 - Physical • Layer 2 - Data link • Layer 3 - Network • Layer 4 - Transport • Layer 5 - Session • Layer 6 - Presentation • Layer 7 - Application	1-4: The Seven-Layer OSI Model	Understand
1.2 Compare and contrast networking appliances, applications, and functions. • Physical and virtual appliances ▪ Router ▪ Switch ▪ Firewall ▪ Intrusion detection system (IDS)/intrusion prevention system (IPS) ▪ Load balancer ▪ Proxy ▪ Network-attached storage (NAS) ▪ Storage area network (SAN) ▪ Wireless ◆ Access point (AP) ◆ Controller • Applications ▪ Content delivery network (CDN) • Functions ▪ Virtual private network (VPN) ▪ Quality of service (QoS) ▪ Time to live (TTL)	1-3: Network Hardware 1-5: Safety Procedures and Policies 3-3: Ports and Sockets 3-4: Domain Names and DNS 4-1: TCP/IP Core Protocols 4-3: Remote Access Protocols 6-1: Characteristics of Wireless Transmissions 6-2: 802.11 WLAN Standards 7-1: Physical Architecture 8-1: Why Segment Networks 8-3: VLANs (Virtual LANs) 8-4: Cloud Networking 9-1: WAN Connectivity 9-2: Routing Protocols 9-3: Securing WAN Connections 10-2: Risk Assessment and Management 12-2: Manage Network Traffic	Understand

Objective	Section	Bloom's Taxonomy
1.3 Summarize cloud concepts and connectivity options. • Network functions virtualization (NFV) • Virtual private cloud (VPC) • Network security groups • Network security lists • Cloud gateways ▪ Internet gateway ▪ Network address translation (NAT) gateway • Cloud connectivity options ▪ VPN ▪ Direct Connect • Deployment models ▪ Public ▪ Private ▪ Hybrid • Service models ▪ Software as a service (SaaS) ▪ Infrastructure as a service (IaaS) ▪ Platform as a service (PaaS) • Scalability • Elasticity • Multitenancy	7-2: Virtual Architecture 7-3: Cloud Architecture 8-4: Cloud Networking 9-1: WAN Connectivity 9-3: Securing WAN Connections	Understand

Objective		Section	Bloom's Taxonomy
1.4 Explain common networking ports, protocols, services, and traffic types. Protocols and Ports		1-2: Network Services 2-1: Components of Structured Cabling 3-2: IP Addresses 3-3: Ports and Sockets 3-4: Domain Names and DNS 3-5: Troubleshooting Address Problems 4-1: TCP/IP Core Protocols 4-2: Encryption Protocols 4-3: Remote Access Protocols 4-4: Troubleshooting Network Issues 11-3: Authentication Technologies 12-1: Collect Network Data	Understand
File Transfer Protocol (FTP)	20/21		
Secure File Transfer Protocol (SFTP)	22		
Secure Shell (SSH)	22		
Telnet	23		
Simple Mail Transfer Protocol (SMTP)	25		
Domain Name System (DNS)	53		
Dynamic Host Configuration Protocol (DHCP)	67/68		
Trivial File Transfer Protocol (TFTP)	69		
Hypertext Transfer Protocol (HTTP)	80		
Network Time Protocol (NTP)	123		
Simple Network Management Protocol (SNMP)	161/162		
Lightweight Directory Access Protocol (LDAP)	389		
Hypertext Transfer Protocol Secure (HTTPS)	443		
Server Message Block (SMB)	445		
Syslog	514		
Simple Mail Transfer Protocol Secure (SMTPS)	587		
Lightweight Directory Access Protocol over SSL (LDAPS)	636		
Structured Query Language (SQL) Server	1433		
Remote Desktop Protocol (RDP)	3389		
Session Initiation Protocol (SIP) • Internet Protocol (IP) types ▪ Internet Control Message Protocol (ICMP) ▪ Transmission Control Protocol (TCP) ▪ User Datagram Protocol (UDP) ▪ Generic Routing Encapsulation (GRE) ▪ Internet Protocol Security (IPSec) ◆ Authentication Header (AH) ◆ Encapsulating Security Payload (ESP) ◆ Internet Key Exchange (IKE) • Traffic types ▪ Unicast ▪ Multicast ▪ Anycast ▪ Broadcast	5060/5061		

Objective	Section	Bloom's Taxonomy
1.5 Compare and contrast transmission media and transceivers. • Wireless ▪ 802.11 standards ▪ Cellular ▪ Satellite • Wired ▪ 802.3 standards ▪ Single-mode vs. multimode fiber ▪ Direct attach copper (DAC) cable ◆ Twinaxial cable ▪ Coaxial cable ▪ Cable speeds ▪ Plenum vs. non-plenum cable • Transceivers ▪ Protocol ◆ Ethernet ◆ Fibre Channel (FC) ▪ Form factors ◆ Small form-factor pluggable (SFP) ◆ Quad small form-factor pluggable (QSFP) • Connector types ▪ Subscriber connector (SC) ▪ Local connector (LC) ▪ Straight tip (ST) ▪ Multi-fiber push on (MPO) ▪ Registered jack (RJ)11 ▪ RJ45 ▪ F-type ▪ Bayonet Neill-Concelman (BNC)	5-2: Copper Cable 5-3: Fiber-Optic Cable 6-2: 802.11 WLAN Standards 8-1: Why Segment Networks 9-1: WAN Connectivity	Understand
1.6 Compare and contrast network topologies, architectures, and types. • Mesh • Hybrid • Star/hub and spoke • Spine and leaf • Point to point • Three-tier hierarchical model ▪ Core ▪ Distribution ▪ Access • Collapsed core • Traffic flows ▪ North-south ▪ East-west	1-3: Network Hardware 7-1: Physical Architecture 9-1: WAN Connectivity	Understand

Objective	Section	Bloom's Taxonomy
1.7 Given a scenario, use appropriate IPv4 network addressing. • Public vs. private ▪ Automatic Private IP Addressing (APIPA) ▪ RFC1918 ▪ Loopback/localhost • Subnetting ▪ Variable Length Subnet Mask (VLSM) ▪ Classless Inter-domain Routing (CIDR) • IPv4 address classes ▪ Class A ▪ Class B ▪ Class C ▪ Class D ▪ Class E	3-1: Addressing Overview 3-2: IP Addresses 8-2: Subnetting	Apply
1.8 Summarize evolving use cases for modern network environments. • Software-defined network (SDN) and software-defined wide area network (SD-WAN) ▪ Application aware ▪ Zero-touch provisioning ▪ Transport agnostic ▪ Central policy management • Virtual Extensible Local Area Network (VXLAN) ▪ Data center interconnection (DCI) ▪ Layer 2 encapsulation • Zero trust architecture (ZTA) ▪ Policy-based authentication ▪ Authorization ▪ Least privilege access • Secure Access Secure Edge (SASE)/Security Service Edge (SSE) • Infrastructure as code (IaC) ▪ Automation ◆ Playbooks/templates/reusable tasks ◆ Configuration drift/compliance ◆ Upgrades ◆ Dynamic inventories ▪ Source control ◆ Version control ◆ Central repository ◆ Conflict identification ◆ Branching • IPv6 addressing ▪ Mitigating address exhaustion ▪ Compatibility requirements ▪ Tunneling ▪ Dual stack ▪ NAT64	2-3: Asset Management 3-1: Addressing Overview 3-2: IP Addresses 7-2: Virtual Architecture 7-4: Automating Networks 8-3: VLANs (Virtual LANs) 9-1: WAN Connectivity 11-4: Modernizing Network Security	Understand

Domain 2.0 Network Implementation—20% of Examination

Objective	Section	Bloom's Taxonomy
2.1 Explain characteristics of routing technologies. • Static routing • Dynamic routing ▪ Border Gateway Protocol (BGP) ▪ Enhanced Interior Gateway Routing Protocol (EIGRP) ▪ Open Shortest Path First (OSPF) • Route selection ▪ Administrative distance ▪ Prefix length ▪ Metric • Address translation ▪ NAT ▪ Port address translation (PAT) • First Hop Redundancy Protocol (FHRP) • Virtual IP (VIP) • Subinterfaces	3-2: IP Addresses 7-1: Physical Architecture 8-3: VLANs (Virtual LANs) 9-2: Routing Protocols	Understand
2.2 Given a scenario, configure switching technologies and features. • Virtual Local Area Network (VLAN) ▪ VLAN database ▪ Switch Virtual Interface (SVI) • Interface configuration ▪ Native VLAN ▪ Voice VLAN ▪ 802.1Q tagging ▪ Link aggregation ▪ Speed ▪ Duplex • Spanning tree • Maximum transmission unit (MTU) ▪ Jumbo frames	4-1: TCP/IP Core Protocols 5-1: Transmission Basics 7-1: Physical Architecture 8-3: VLANs (Virtual LANs) 9-4: Troubleshooting WAN Connections	Apply

Objective	Section	Bloom's Taxonomy
2.3 Given a scenario, select and configure wireless devices and technologies. • Channels ▪ Channel width ▪ Non-overlapping channels ▪ Regulatory impacts ◆ 802.11h • Frequency options ▪ 2.4 GHz ▪ 5 GHz ▪ 6 GHz ▪ Band steering • Service set identifier (SSID) ▪ Basic service set identifier (BSSID) ▪ Extended service set identifier (ESSID) • Network types ▪ Mesh networks ▪ Ad hoc ▪ Point to point ▪ Infrastructure • Encryption ▪ Wi-Fi Protected Access 2 (WPA2) ▪ WPA3 • Guest networks ▪ Captive portals • Authentication ▪ Pre-shared key (PSK) vs. Enterprise • Antennas ▪ Omnidirectional vs. directional • Autonomous vs. lightweight access point	6-1: Characteristics of Wireless Transmissions 6-2: 802.11 WLAN Standards 6-4: Wi-Fi Network Security	Apply
2.4 Explain important factors of physical installations. • Important installation implications ▪ Locations ◆ Intermediate distribution frame (IDF) ◆ Main distribution frame (MDF) ▪ Rack size ▪ Port-side exhaust/intake ▪ Cabling ◆ Patch panel ◆ Fiber distribution panel ▪ Lockable • Power ▪ Uninterruptible power supply (UPS) ▪ Power distribution unit (PDU) ▪ Power load ▪ Voltage • Environmental factors ▪ Humidity ▪ Fire suppression ▪ Temperature	1-5: Safety Procedures and Policies 2-1: Components of Structured Cabling 5-3: Fiber-Optic Cable 11-1: Security and Users 12-1: Collect Network Data 12-3: Plan Response and Recovery Strategies	Understand

Domain 3.0 Network Operations—19% of Examination

Objective	Section	Bloom's Taxonomy
3.1 Explain the purpose of organizational processes and procedures. • Documentation ▪ Physical vs. logical diagrams ▪ Rack diagrams ▪ Cable maps and diagrams ▪ Network diagrams ◆ Layer 1 ◆ Layer 2 ◆ Layer 3 ▪ Asset inventory ◆ Hardware ◆ Software ◆ Licensing ◆ Warranty support ▪ IP address management (IPAM) ▪ Service-level agreement (SLA) ▪ Wireless survey/heat map • Life-cycle management ▪ End-of-life (EOL) ▪ End-of-support (EOS) ▪ Software management ◆ Patches and bug fixes ◆ Operating system (OS) ◆ Firmware ▪ Decommissioning • Change management ▪ Request process tracking/service request • Configuration management ▪ Production configuration ▪ Backup configuration ▪ Baseline/golden configuration	1-1: Network Models 1-6: Troubleshooting Network Problems 2-2: Network Documentation 2-3: Asset Management 3-2: IP Addresses 4-3: Remote Access Protocols 5-2: Copper Cable 5-4: Cable Troubleshooting Tools 6-3: Implementing a Wi-Fi Network 10-3: Device Hardening 12-3: Plan Response and Recovery Strategies	Understand

Objective	Section	Bloom's Taxonomy
3.2 Given a scenario, use network monitoring technologies. • Methods ▪ SNMP ◆ Traps ◆ Management information base (MIB) ◆ Versions ○ v2c ○ v3 ◆ Community strings ◆ Authentication ▪ Flow data ▪ Packet capture ▪ Baseline metrics ◆ Anomaly alerting/notification ▪ Log aggregation ◆ Syslog collector ◆ Security information and event management (SIEM) ▪ Application programming interface (API) integration ▪ Port mirroring • Solutions ▪ Network discovery ◆ Ad hoc ◆ Scheduled ▪ Traffic analysis ▪ Performance monitoring ▪ Availability monitoring ▪ Configuration monitoring	2-3: Asset Management 4-4: Troubleshooting Network Issues 10-2: Risk Assessment and Management 11-2: AAA (Authentication, Authorization, and Accounting) 12-1: Collect Network Data 12-2: Manage Network Traffic	Apply
3.3 Explain disaster recovery (DR) concepts. • DR metrics ▪ Recovery point objective (RPO) ▪ Recovery time objective (RTO) ▪ Mean time to repair (MTTR) ▪ Mean time between failures (MTBF) • DR sites ▪ Cold site ▪ Warm site ▪ Hot site • High-availability approaches ▪ Active-active ▪ Active-passive • Testing ▪ Tabletop exercises ▪ Validation tests	2-3: Asset Management 9-2: Routing Protocols 12-3: Plan Response and Recovery Strategies	Understand

Objective	Section	Bloom's Taxonomy
3.4 Given a scenario, implement IPv4 and IPv6 network services. • Dynamic addressing ▪ DHCP ◆ Reservations ◆ Scope ◆ Lease time ◆ Options ◆ Relay/IP helper ◆ Exclusions ▪ Stateless address autoconfiguration (SLAAC) • Name resolution ▪ DNS ◆ Domain Name Security Extensions (DNSSEC) ◆ DNS over HTTPS (DoH) and DNS over TLS (DoT) ◆ Record types ○ Address (A) ○ AAAA ○ Canonical name (CNAME) ○ Mail exchange (MX) ○ Text (TXT) ○ Nameserver (NS) ○ Pointer (PTR) ◆ Zone types ○ Forward ○ Reverse ◆ Authoritative vs. non-authoritative ◆ Primary vs. secondary ◆ Recursive ▪ Hosts file • Time protocols ▪ NTP ▪ Precision Time Protocol (PTP) ▪ Network Time Security (NTS)	3-2: IP Addresses 3-3: Ports and Sockets 3-4: Domain Names and DNS 3-5: Troubleshooting Address Problems 4-2: Encryption Protocols 8-2: Subnetting 11-4: Modernizing Network Security	Apply
3.5 Compare and contrast network access and management methods. • Site-to-site VPN • Client-to-site VPN ▪ Clientless ▪ Split tunnel vs. full tunnel • Connection methods ▪ SSH ▪ Graphical user interface (GUI) ▪ API ▪ Console • Jump box/host • In-band vs. out-of-band management	1-4: The Seven-Layer OSI Model 2-1: Components of Structured Cabling 4-3: Remote Access Protocols 7-3: Cloud Architecture 8-4: Cloud Networking	Understand

Domain 4.0 Network Security—14% of Examination

Objective	Section	Bloom's Taxonomy
4.1 Explain the importance of basic network security concepts. • Logical security ▪ Encryption ◆ Data in transit ◆ Data at rest ▪ Certificates ◆ Public key infrastructure (PKI) ◆ Self-signed ▪ Identity and access management (IAM) ◆ Authentication ○ Multifactor authentication (MFA) ○ Single sign-on (SSO) ○ Remote Authentication Dial-in User Service (RADIUS) ○ LDAP ○ Security Assertion Markup Language (SAML) ○ Terminal Access Controller Access Control System Plus (TACACS+) ○ Time-based authentication ◆ Authorization ○ Least privilege ○ Role-based access control ▪ Geofencing • Physical security ▪ Camera ▪ Locks • Deception technologies ▪ Honeypot ▪ Honeynet • Common security terminology ▪ Risk ▪ Vulnerability ▪ Exploit ▪ Threat ▪ Confidentiality, Integrity, and Availability (CIA) triad • Audits and regulatory compliance ▪ Data locality ▪ Payment Card Industry Data Security Standards (PCI DSS) ▪ General Data Protection Regulation (GDPR) • Network segmentation enforcement ▪ Internet of Things (IoT) and Industrial Internet of Things (IIoT) ▪ Supervisory control and data acquisition (SCADA), industrial control system (ICS), operational technology (OT) ▪ Guest ▪ Bring your own device (BYOD)	2-1: Components of Structured Cabling 4-2: Encryption Protocols 6-3: Implementing a Wi-Fi Network 6-4: Wi-Fi Network Security 8-1: Why Segment Networks 10-1: Security Risks 10-2: Risk Assessment and Management 11-1: Security and Users 11-2: AAA (Authentication, Authorization, and Accounting) 11-3: Authentication Technologies	Understand

Objective	Section	Bloom's Taxonomy
4.2 Summarize various types of attacks and their impact to the network. • Denial-of-service (DoS)/distributed denial-of-service (DDoS) • VLAN hopping • Media Access Control (MAC) flooding • Address Resolution Protocol (ARP) poisoning • ARP spoofing • DNS poisoning • DNS spoofing • Rogue devices and services ▪ DHCP ▪ AP • Evil twin • On-path attack • Social engineering ▪ Phishing ▪ Dumpster diving ▪ Shoulder surfing ▪ Tailgating • Malware	6-4: Wi-Fi Network Security 7-1: Physical Architecture 8-3: VLANs (Virtual LANs) 10-1: Security Risks 10-3: Device Hardening	Understand
4.3 Given a scenario, apply network security features, defense techniques, and solutions. • Device hardening ▪ Disable unused ports and services ▪ Change default passwords • Network access control (NAC) ▪ Port security ▪ 802.1X ▪ MAC filtering • Key management • Security rules ▪ Access control list (ACL) ▪ Uniform Resource Locator (URL) filtering ▪ Content filtering • Zones ▪ Trusted vs. untrusted ▪ Screened subnet	4-2: Encryption Protocols 6-4: Wi-Fi Network Security 7-1: Physical Architecture 8-1: Why Segment Networks 9-3: Securing WAN Connections 10-3: Device Hardening 11-2: AAA (Authentication, Authorization, and Accounting) 11-4: Modernizing Network Security	Apply

Domain 5.0 Network Troubleshooting—24% of Examination

Objective	Section	Bloom's Taxonomy
5.1 Explain the troubleshooting methodology. • Identify the problem ▪ Gather information ▪ Question users ▪ Identify symptoms ▪ Determine if anything has changed ▪ Duplicate the problem, if possible ▪ Approach multiple problems individually • Establish a theory of probable cause ▪ Question the obvious ▪ Consider multiple approaches ◆ Top-to-bottom/bottom-to-top OSI model ◆ Divide and conquer • Test the theory to determine the cause ▪ If theory is confirmed, determine next steps to resolve problem ▪ If theory is not confirmed, establish a new theory or escalate • Establish a plan of action to resolve the problem and identify potential effects • Implement the solution or escalate as necessary • Verify full system functionality and implement preventive measures if applicable • Document findings, actions, outcomes, and lessons learned throughout the process	1-6: Troubleshooting Network Problems	Understand

Objective	Section	Bloom's Taxonomy
5.2 Given a scenario, troubleshoot common cabling and physical interface issues. • Cable issues ▪ Incorrect cable ◆ Single mode vs. multimode ◆ Category 5/6/7/8 ◆ Shielded twisted pair (STP) vs. unshielded twisted pair (UTP) ▪ Signal degradation ◆ Crosstalk ◆ Interference ◆ Attenuation ▪ Improper termination ▪ Transmitter (TX)/Receiver (RX) transposed • Interface issues ▪ Increasing interface counters ◆ Cyclic redundancy check (CRC) ◆ Runts ◆ Giants ◆ Drops ▪ Port status ◆ Error disabled ◆ Administratively down ◆ Suspended • Hardware issues ▪ Power over Ethernet (PoE) ◆ Power budget exceeded ◆ Incorrect standard ▪ Transceivers ◆ Mismatch ◆ Signal strength	5-1: Transmission Basics 5-2: Copper Cable 5-3: Fiber-Optic Cable 5-4: Cable Troubleshooting Tools 7-1: Physical Architecture 9-4: Troubleshooting WAN Connections 12-1: Collect Network Data 12-2: Manage Network Traffic	Analyze
5.3 Given a scenario, troubleshoot common issues with network services. • Switching issues ▪ STP ◆ Network loops ◆ Root bridge selection ◆ Port roles ◆ Port states ▪ Incorrect VLAN assignment ▪ ACLs • Route selection ▪ Routing table ▪ Default routes • Address pool exhaustion • Incorrect default gateway • Incorrect IP address ▪ Duplicate IP address • Incorrect subnet mask	3-5: Troubleshooting Address Problems 7-1: Physical Architecture 8-2: Subnetting 8-3: VLANs (Virtual LANs) 9-3: Securing WAN Connections 9-4: Troubleshooting WAN Connections	Analyze

Objective	Section	Bloom's Taxonomy
5.4 Given a scenario, troubleshoot common performance issues. • Congestion/contention • Bottlenecking • Bandwidth ▪ Throughput capacity • Latency • Packet loss • Jitter • Wireless ▪ Interference ◆ Channel overlap ▪ Signal degradation or loss ▪ Insufficient wireless coverage ▪ Client disassociation issues ▪ Roaming misconfiguration	5-1: Transmission Basics 6-5: Troubleshooting Wi-Fi Networks 9-4: Troubleshooting WAN Connections 12-1: Collect Network Data 12-2: Manage Network Traffic	Analyze
5.5 Given a scenario, use the appropriate tool or protocol to solve network issues. • Software tools ▪ Protocol analyzer ▪ Command line ◆ ping ◆ traceroute/tracert ◆ nslookup ◆ tcpdump ◆ dig ◆ netstat ◆ ip/ifconfig/ipconfig ◆ arp ▪ Nmap ▪ Link Layer Discovery Protocol (LLDP)/Cisco Discovery Protocol (CDP) ▪ Speed tester • Hardware tools ▪ Toner ▪ Cable tester ▪ Taps ▪ Wi-Fi analyzer ▪ Visual fault locator • Basic network device commands ▪ show mac-address-table ▪ show route ▪ show interface ▪ show config ▪ show arp ▪ show vlan ▪ show power	2-2: Network Documentation 3-1: Addressing Overview 3-5: Troubleshooting Address Problems 4-1: TCP/IP Core Protocols 4-4: Troubleshooting Network Issues 5-4: Cable Troubleshooting Tools 6-5: Troubleshooting Wi-Fi Networks 8-3: VLANs (Virtual LANs) 9-1: WAN Connectivity 9-2: Routing Protocols 9-4: Troubleshooting WAN Connections 10-2: Risk Assessment and Management 12-3: Plan Response and Recovery Strategies	Apply